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Method for Forming a Semiconductor Device, and a Semiconductor Device

Abstract

A method and a semiconductor device are provided. An example method may include forming a stack of field effect transistors, FETs, on top of a substrate. The method may also include forming a first insulating layer laterally surrounding the stack of FETs and etching a hole into the first insulating layer, the hole extending between a top endpoint and a bottom endpoint. The method may further include filling the hole with electrically conductive material such that the electrically conductive material of the filled hole forms a via. Moreover, the method may include etching, from a bottom side of the substrate towards the first insulating layer, a trench, a top part of the trench comprising both a bottom side of the first source/drain region and the bottom endpoint. The method may also include filling the top part of the trench with electrically conductive material.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a non-provisional patent application claiming priority to application number EP 24160444.6, filed on Feb. 29, 2024, the contents of which are hereby incorporated by reference.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to a semiconductor device and a method for forming the semiconductor device.

BACKGROUND

[0003] To enable more area- and power-efficient circuits, vertical semiconductor device structures are receiving increasing attention as an alternative to traditional planar semiconductor devices. One notable example is stacked transistor devices comprising a complementary pair of field effect transistors (FETs) stacked on top of each other, i.e. a p-type FET (pFET) on top of a n-type FET (nFET) or vice versa. A complementary pair of FETs stacked on top of each other is called a complementary FET (CFET).

SUMMARY

[0004] The disclosure broadly relates to methods and devices that may facilitate a compact semiconductor device, in particular, a compact CFET.

[0005] The disclosure also relates to methods and devices that may facilitate easy and/or low cost manufacturing of the semiconductor device. For example, facilitating relaxed constraints for the thermal budget during manufacturing and/or relaxed constraints for contamination.

[0006] The disclosure further relates to methods and devices that may enable a high quality semiconductor device. For example, disclosed devices may include reduced thermal damage to metal contacts and dielectric layers. Another example is devices with reduced metal contamination.

[0007] Aspects of the disclosure are at least partly described in the independent claims. Other embodiments are set out in the dependent claims. These and other aspects of the disclosure will be apparent to one of skill in the art.

[0008] According to a first aspect, there is provided a method for forming a semiconductor device, the method includes: forming a stack of field effect transistors, FETs, on top of a substrate, the stack of FETs including a bottom FET and a top FET, the top FET being stacked on top of the bottom FET, the bottom FET comprising at least a first source/drain region; forming a first insulating layer, the first insulating layer laterally surrounding the stack of FETs; etching a hole into the first insulating layer, the hole being laterally spaced apart from the stack of FETs, the hole extending between a top endpoint and a bottom endpoint, the top endpoint being arranged at a top side of the first insulating layer, the bottom endpoint being arranged at a level below a bottom level of the top FET; filling the hole with electrically conductive material such that the electrically conductive material of the filled hole forms a via; etching, from a bottom side of the substrate towards the first insulating layer, a trench, such that a top part of the trench comprises both a bottom side of the first source/drain region and the bottom endpoint; and filling the top part of the trench with electrically conductive material, such that the electrically conductive material of the filled top part of the trench electrically connects the bottom side of the first source/drain region to the bottom endpoint of the via, whereby the filled top part of the trench forms an interconnect.

[0009] Relative spatial terms such as “top”, “bottom”, “lower”, “vertical”, “stacked on top of”, are herein to may mean denoting locations or directions within a frame of reference of the semiconductor device. In particular, the terms may be understood in relation to a normal direction to the substrate on which the device layer stack may be formed, or equivalently in relation to a bottom-up direction of the device layer stack. Correspondingly, terms such as “lateral” and “horizontal” are may mean locations or directions parallel to the substrate.

[0010] The bottom FET may be a pFET and the top FET may be an nFET, or vice versa. Thus, the stack of FETs may be a CFET. Alternatively, both the bottom FET and the top FET may be pFETs. Alternatively, both the bottom FET and the top FET may be nFETs.

[0011] Each of the bottom FET and the top FET may comprise at least one channel layer for charge transport. The channel layers may be part of a stack of layers forming a fin. The fin may comprise two opposing lateral side faces, two opposing lateral end faces, and a top face. A bottom face of the fin may be facing the substrate. Each channel layer may comprise a semiconductor, e.g. silicon. The substrate may comprise a semiconductor. The substrate may be e.g. a silicon substrate or a silicon-on-insulator substrate.

[0012] Each of the bottom FET and the top FET may comprise two source/drain regions, being arranged at opposite ends of the at least one channel layer, i.e. at the opposing lateral end faces of the fin. Thus, the first source/drain region may be a source/drain region at one of the ends of the channel layer of the bottom FET. Each source/drain region may comprise a semiconductor, e.g. silicon. Each source/drain region may be doped, e.g. p doped when belonging to a pFET or n doped when belonging to an nFET.

[0013] Each of the bottom FET and the top FET may comprise a gate configured to control the charge transport through the at least one channel layer. The gate may be e.g. a gate-all-around extending around the channel layers or a tri-gate at three sides of the channel layers.

[0014] The first insulating layer may be called zero-level interlayer dielectric (ILDO). The first insulating layer may comprise electrically insulating material, e.g. dielectric material. The first insulating layer may comprise SiO.sub.2 and/or SiOC. The first insulating layer may be formed by e.g. Plasma-Enhanced Chemical Vapor Deposition (PECVD) or Low-Pressure Chemical Vapor Deposition (LPCVD).

[0015] The via and the interconnect may form an electrical connection from the first source/drain region to the top side of the first insulating layer. Thus, the interconnect may be an electrical connection from the first source/drain region to the via and the via may be an electrical connection from the interconnect to the top side of the first insulating layer.

[0016] The interconnect between the first source/drain region and the via may facilitate a compact semiconductor device. The interconnect may extend laterally from the first source/drain region to the via. Thus, the interconnect and the via may form a short and/or compact signal line to the top side of the first insulating layer. Thus, there may be no need to rout the signal line to a backside power delivery network and then to dedicated vias between the backside power delivery network and the top side of the first insulating layer. Such routing through a backside power delivery network may be associated with an area penalty.

[0017] According to example embodiments, the trench may be etched from the bottom side of the substrate, i.e. by backside etching. Thus, at least part of the process of forming the interconnect may be performed from the bottom side of the substrate. For example, filling the top part of the trench with electrically conductive material may be performed from the bottom side of the substrate.

[0018] Bottom side processing (i.e. backside processing) may facilitate relaxed constraints for the thermal budget during manufacturing. As an example, lateral interconnects grown epitaxially may not be needed. Thus, the stack of FETs may not need to be subjected to the heat associated with such growth. In other words, the stack of FETs may not need to be heated unnecessarily, which may reduce heat damage to metal contacts and dielectric layers.

[0019] Bottom side processing (i.e. back side processing) may facilitate relaxed constraints for contamination. For example, metal contacts and/or metal interconnects may be formed after epitaxial growth, e.g. after epitaxial growth of source/drain regions. Thus, it may be possible to avoid metal going into the epitaxy machine. Such metal may contaminate the epitaxy machine and may affect the growth run in question or future growth runs.

[0020] As mentioned, the trench may be etched from the bottom side of the substrate, this may be called backside trench-etching.

[0021] The hole may be etched from the top side of the first insulating layer, this may be called frontside hole-etching.

[0022] Alternatively, the hole may be etched from the bottom side of the substrate, this may be called backside hole-etching.

[0023] Frontside and backside hole-etching will be discussed further below.

[0024] Etching may be performed by dry etching, e.g. by plasma etching. Alternatively or additionally, etching may be performed by wet etching.

[0025] Etching from the bottom side of the substrate may be construed as the bottom side of the substrate facing the etchant and/or being exposed to the etchant, e.g. the bottom side of the substrate facing a plasma etchant. Analogously, etching from the top side of the first insulating layer may be construed as the top side of the first insulating layer facing the etchant and/or being exposed to the etchant, e.g. the top side of the first insulating layer facing a plasma etchant.

[0026] As mentioned, the hole may be etched into the first insulating layer, laterally spaced apart from the stack of FETs. Thus, the hole may be laterally spaced apart from source/drain regions of the stack of FETs, e.g. by at least 5 nm, or at least 10 nm. Thus, the hole may be laterally spaced apart from the channel layers of the stack of FETs. Thus, the hole may be laterally spaced apart from the gates of the stack of FETs.

[0027] Etching the trench from the bottom side of the substrate may be performed after thinning at least parts of the substrate from the bottom side.

[0028] A position of an etched hole or trench may be defined by patterning, e.g. lithographic patterning.

[0029] It should be understood that one or more etch stop layers may be used to stop the etch at desired levels.

[0030] Filling may be performed by deposition, e.g. Chemical Vapor Deposition (CVD), Physical Vapor Deposition (PVD), or Atomic Layer Deposition (ALD). The electrically conductive material filling the hole may be e.g. tungsten (W), molybdenum (Mo), ruthenium (Ru), cobalt (Co), with or without an adhesion liner, e.g., titanium nitride (TiN). The electrically conductive material filling the top part of the trench may be e.g. tungsten (W), molybdenum (Mo), ruthenium (Ru), cobalt (Co), with or without an adhesion liner, e.g., titanium nitride (TiN). In particular, the electrically conductive material filling the hole and/or filling the top part of the trench may be a metal.

[0031] Filling from the bottom side of the substrate may be construed as the bottom side of the substrate facing the deposition source, e.g. the bottom side of the substrate facing a CVD source, a PVD source or an ALD source. Analogously, filling from the top side of the first insulating layer may be construed as the top side of the first insulating layer facing the deposition source.

[0032] The top part of the trench may comprise also at least part of a lateral side of the first source/drain region, such that the electrically conductive material of the filled top part of the trench electrically connects also the lateral side of the first source/drain region to the via.

[0033] Thus, the trench may be etched past the bottom side of the first source/drain region, thereby exposing the bottom side of the first source/drain region as well as part of a lateral side of the first source/drain region. Thus, the bottom side of the first source/drain region as well as part of a lateral side of the first source/drain region may be electrically connected by the electrically conductive material filling the top part of the trench.

[0034] When the electrically conductive material of the filled top part of the trench electrically connects both the bottom side of the first source/drain region and the lateral side of the first source/drain region to the via, the resistance of the interconnect may be small. For example, a large

contact area between the electrically conductive material of the filled top part of the trench and the first source/drain region may be achieved when the electrically conductive material of the filled top part connects both the bottom side of the first source/drain region and the lateral side of the first source/drain region to the via. Thus, a contact resistance may be small.
[0035] The method may enable a small vertical distance between the bottom FET and the top FET. A vertical distance between the bottom FET and the top FET may be smaller than 50 nm. In other words, a vertical distance between the uppermost channel layer of the bottom FET and the lowermost channel layer of the top FET may be smaller than 50 nm. In contrast, an interconnect formed by front side processing may need a larger vertical distance between the bottom FET and the top FET to avoid a short circuit between the bottom FET and the top FET. Depending on the precision needed, other values for the vertical distance may be foreseen. A vertical distance between the bottom FET and the top FET may be smaller than 70 nm. A vertical distance between the bottom FET and the top FET may be smaller than 30 nm.
[0036] Another aspect of the disclosure is a method. The method may further comprise: forming a second insulating layer below the electrically conductive material of the filled top part of the trench; and forming a backside power delivery network below the second insulating layer, wherein the second insulating layer electrically insulates the backside power delivery network from the electrically conducive material of the filled top part of the trench.
[0037] The backside power delivery network may be used for routing of signals. Since the interconnect and the via may rout a signal from the first source/drain region to the top side of the first insulating layer without passing through the backside power delivery network, the space below the interconnect is free for routing of other signals (e.g. from other stacks of FETs) in the backside power delivery network. The second insulating layer may thus electrically insulate the interconnect from the backside power delivery network. The second insulating layer may comprise SiO.sub.2 and/or SiOC. The second insulating layer may be formed by e.g. Plasma-Enhanced Chemical Vapor Deposition (PECVD) or Low-Pressure Chemical Vapor Deposition (LPCVD).
[0038] The electrically conductive material filling the top part of the trench may be a metal.
[0039] Thus, epitaxially grown lateral connections may be avoided. The metal filling the top part of the trench may be deposited from the bottom side of the substrate.
[0040] Forming the stack of FETs may comprise: forming source/drain regions of the bottom and top FET by epitaxial growth, wherein filling the hole with electrically conductive material and filling the top part of the trench with electrically conductive material are performed after forming the source/drain regions of the bottom and top FET.
[0041] Thus, the electrically conductive material of the hole and the electrically conductive material of the top part of the trench may be prevented from contaminating the epitaxy machine.
[0042] The hole may be etched in various ways, sorted into two major categories which may be called frontside hole-etching and backside hole-etching, as mentioned above.
[0043] Etching the hole may comprise etching at least part of the hole from a top side of the first insulating layer. This may be seen as frontside hole-etching. Etching part of the hole from a top side of the first insulating layer may be compatible with methods for forming vias through an ILDO layer. Further, patterning of the hole may be easy from the frontside. Thus, frontside hole-etching may be easy and/or inexpensive to implement.
[0044] An etchstop, also called etch stop layer, may be used to stop the etch at a particular level. For example, frontside hole-etching may be performed with an etchstop. For example, the first insulating layer may comprise a first sublayer, a second sublayer stacked on top of the first sublayer, and a third sublayer stacked on top of the second sublayer, the second sublayer being arranged between the bottom and top FETs, the second and third sublayers comprising different insulating materials; forming the hole may then comprise: selectively etching a first part of the hole through the third sublayer down to the second sublayer; and forming a second part of the hole by expanding the first part of the hole through the second sublayer.

[0045] Etching the first part of the hole may be performed by a first etch. The first etch may be configured to have a high etch rate for the third sublayer and a low etch rate for the second sublayer. The etch rate for the third sublayer and the second sublayer may differ by at least a factor **10**.
[0046] Thus, the first etch may stop or slow down substantially when reaching the second sublayer. In other words, it may be possible to ensure that the etch has reached a level between the bottom and top FETs by etching for a long time while the second sublayer (acting as an etchstop) prevents the etch from going too far.
[0047] Forming a second part of the hole by expanding the first part of the hole through the second sublayer may then be performed using a second etch. Forming the second part of the hole may be called punch-through. The second etch may be different from the first etch. The second etch may be non-selective. The second etch may have a higher etch rate for the material of the second sublayer than the first etch.
[0048] The hole may be filled with electrically conductive material from the top side of the first insulating layer, this may be called frontside hole-filling. Alternatively, the hole may be filled with electrically conductive material from the bottom side of the substrate, after forming the trench, this may be called backside hole-filling.
[0049] Frontside hole-filling may be easy and/or inexpensive to implement.
[0050] Backside hole-filling may make it possible to perform both two acts of filling the hole with electrically conductive material and filling the top part of the trench with electrically conductive material, from the bottom side of the substrate. Thus, the hole and the top part of the trench may be filled with electrically conductive material during one single pass through the deposition machine. There may be no need to first fill the hole from the top side of the first insulating layer, in a first pass through the deposition machine; then take the semiconductor device out of the deposition machine; and later fill the top part of the trench from the bottom side of the substrate in a second pass through the deposition machine. Further, good electrical contact between the interconnect and the via may be achieved.
[0051] Backside hole-filling may be implemented together with frontside hole-etching, e.g. by the use of sacrificial material. For example, frontside hole-etching may be followed by deposition of sacrificial material. For example, the method may comprise: depositing a sacrificial material in the hole before etching the trench; and removing the sacrificial material in the hole after etching the trench, by etching, from the bottom side of the substrate, the sacrificial material, wherein filling the hole with electrically conductive material is performed by filling, from the bottom side of the substrate, the hole with electrically conductive material after removing the sacrificial material.
[0052] Thus, the hole may be etched from the top side of the first insulating layer. The sacrificial material may then be deposited in the hole. The sacrificial material may be silicon oxide (SiO.sub.2), amorphous silicon (a-Si), or amorphous silicon-germanium (a-SiGe). The trench may subsequently be etched from the bottom side of the substrate until the top part of the trench reaches the bottom endpoint of the sacrificial material. The sacrificial material may then be etched from the bottom side of the substrate until removed. Electrically conductive material may then be deposited, e.g. simultaneously or sequentially, from the bottom side of the substrate, into both the hole and the trench. As understood from the above, filling the top part of the trench with electrically conductive material may be performed from the bottom side of the substrate after filling the hole with electrically conductive material.
[0053] Backside hole-filling may be implemented together with backside hole-etching. For example, etching the hole may be performed after etching the trench and the method may comprise: etching the hole from the bottom side of the substrate, wherein filling the hole with electrically conductive material is performed from the bottom side of the substrate.
[0054] According to another aspect, a semiconductor device is provided.
[0055] According to the another aspect, there is provided a semiconductor device comprising: a

stack of field effect transistors, FETs, arranged on top of a substrate, the stack of FETs comprising a bottom FET and a top FET, the top FET being stacked on top of the bottom FET, the bottom FET comprising at least a first source/drain region; a first insulating layer, the first insulating layer laterally surrounding the stack of FETs; a via comprising a hole filled with electrically conductive material, the hole being laterally spaced apart from the stack of FETs, the hole extending between a top endpoint and a bottom endpoint, the top endpoint being arranged at a top side of the first insulating layer, the bottom endpoint being arranged at a level below a bottom level of the top FET; and an interconnect comprising a top part of a trench filled with electrically conductive material, the trench extending from a bottom side of the substrate towards the first insulating layer, wherein the top part of the trench comprises both a bottom side of the first source/drain region and the bottom endpoint of the via, the electrically conductive material of the filled top part of the trench electrically connecting the bottom side of the first source/drain region to the bottom endpoint of the via.
[0056] The semiconductor device may be configured such that the top part of the trench comprises also at least part of a lateral side of the first source/drain region, such that the electrically conductive material of the filled top part of the trench electrically connects also the lateral side of the first source/drain region to the via.
[0057] The semiconductor device may be configured such that a vertical distance between the bottom FET and the top FET is smaller than 50 nm.
[0058] The semiconductor device may further comprise: a second insulating layer below the electrically conductive material of the filled top part of the trench; and a backside power delivery network below the second insulating layer, wherein the second insulating layer electrically insulates the backside power delivery network from the electrically conductive material of the filled top part of the trench.
[0059] The semiconductor device may be configured such that the electrically conductive material filling the top part of the trench is a metal.

Description

BRIEF DESCRIPTION OF THE FIGURES
[0060] The above, as well as additional, features will be better understood through the following illustrative and non-limiting detailed description of example embodiments, with reference to the appended drawings. In the drawings, like reference numerals will be used for like elements unless stated otherwise.
[0061] FIG. **1** illustrates a cross-sectional view of a semiconductor device, according to an example embodiment.
[0062] FIG. **2**(*a*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0063] FIG. **2**(*b*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0064] FIG. **2**(*c*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0065] FIG. **2**(*d*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0066] FIG. **3**(*a*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0067] FIG. **3**(*b*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0068] FIG. **3**(*c*) illustrates a step of a method for forming a semiconductor device, according to an

example embodiment.
[0069] FIG. **3**(*d*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0070] FIG. **4**(*a*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0071] FIG. **4**(*b*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0072] FIG. **4**(*c*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0073] FIG. **4**(*d*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0074] FIG. **5**(*a*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0075] FIG. **5**(*b*) illustrates a step of a method for forming a semiconductor device, according to an example embodiment.
[0076] All the figures are schematic, not necessarily to scale, and generally only show parts which are necessary to elucidate example embodiments, wherein other parts may be omitted or merely suggested.
DETAILED DESCRIPTION
[0077] Example embodiments will now be described more fully hereinafter with reference to the accompanying drawings. That which is encompassed by the claims may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided by way of example. Furthermore, like numbers refer to the same or similar elements or components throughout.
[0078] In cooperation with attached drawings, the technical contents and detailed description of the present disclosure are described thereinafter according to an example embodiment, being not used to limit the claimed scope. This disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided for thoroughness and completeness.
[0079] The figures illustrate a semiconductor device **1** and a method for forming the semiconductor device **1**.
[0080] Axes X, Y and Z are indicated in all figures. Axes X, Y and Z indicate a first direction, a second direction transverse to the first direction, and a vertical or bottom-up direction, respectively. The X- and Y-direction may in particular be referred to as lateral or horizontal directions in that they are parallel to a main plane of a substrate **4** of the semiconductor device **1**. The Z-direction is parallel to a normal direction to the substrate **4**.
[0081] FIG. **1** illustrates a cross-sectional view of the semiconductor device **1** comprising a stack of FETs **60**, arranged on top of a substrate **4**. The stack of FETs **60** comprising a bottom FET **61** and a top FET **62**, the top FET **62** being stacked on top of the bottom FET **61**.
[0082] The bottom FET **61** may be a pFET and the top FET **62** may be an nFET, or vice versa. Thus, the stack of FETs **60** may be a CFET. Alternatively, both the bottom FET **61** and the top FET **62** may be pFETs. Alternatively, both the bottom FET and the top FET may be nFETs.
[0083] Each of the bottom FET **61** and the top FET **62** may comprise at least one channel layer **55** for charge transport. Each channel layer **55** may comprise a semiconductor, e.g. silicon. The channel layers **55** may be part of a stack of layers forming a fin. The bottom FET **61** may comprise a plurality of channel layers **55**, e.g. at least two channel layers **55** or at least three channel layers **55**. Similarly, the top FET **62** may comprise a plurality of channel layers **55**, e.g. at least two channel layers **55** or at least three channel layers **55**.
[0084] As illustrated, the bottom FET **61** comprises a first source/drain region **51**. The first source/drain region **51** may be arranged on a lateral end face of the fin, in contact with the one or

more channel layer **55** of the bottom FET **61**. The lateral end face of the fin may be orthogonal to the second direction, i.e. Y-direction. FIG. **1** illustrates a cross-sectional view through the first source/drain region **51** but, for clarity, the channel layers **55** of the bottom FET **61** behind the first source/drain region **51** are illustrated with dashed lines. The channel layers **55** of the bottom FET **61** may then extend in the second direction, i.e. Y-direction, to a source/drain region opposite to the first source/drain region **51**. In other words, in the illustrated semiconductor device, current of the bottom FET **61** may flow in the second direction, i.e. Y-direction.

[0085] As illustrated, the top FET **62** comprises a second source/drain region **52**. The second source/drain region **52** may be arranged on the same lateral end face of the fin as the first source/drain region **51**. The top FET **62** may, analogously to the bottom FET **61** have one or more channel layer **55** extending in the second direction between the second source/drain region **52** and a source/drain region opposite to the second source/drain region **52**. The channel layers **55** of the top FET **62**, behind the second source/drain region **52**, are illustrated with dashed lines.

[0086] Each source/drain **51**, **52** region may comprise a semiconductor, e.g. silicon. Each source/drain region **51**, **52** may be doped, e.g. p doped when belonging to a pFET or n doped when belonging to an nFET.

[0087] Each of the bottom FET **61** and the top FET **62** may comprise a gate (not shown) configured to control the charge transport through the at least one channel layer **55**.

[0088] The semiconductor device **1** of FIG. **1** further comprises a first insulating layer **11**, a via **21** and an interconnect **31**. The via **21** and the interconnect comprise electrically conductive material, e.g. metal.

[0089] The first insulating layer **11** may laterally surround the stack of FETs **60**. The first insulating layer **11** may be an ILDO layer. The first insulating layer **11** may comprise SiO.sub.2 and/or SiOC.

[0090] The bottom FET **61** and the top FET **62** may be separated by electrically insulating material. For example, channel layers **55** of the bottom FET **61** may be separated from channel layers **55** of the top FET **62** by electrically insulating material. The first source/drain region **51** may be separated from the second source/drain region **52** by electrically insulating material. Electrically insulating material separating the bottom FET **61** and the top FET **62** may comprise parts of the first insulating layer **11**.

[0091] A vertical distance between the bottom FET **61** and the top FET **62** may be smaller than 50 nm. In other words, a vertical distance between the uppermost channel layer **55** of the bottom FET **61** and the lowermost channel layer **55** of the top FET **62** may be smaller than 50 nm.

[0092] The via **21** may be a hole **20** filled with electrically conductive material. The via **21** may extend from a top endpoint **26**, at a top side **11***t* of the first insulating layer **11**, to a bottom endpoint **25** at a level below a bottom level **62***b* of the top FET **62**. Thus, a vertical distance between the substrate **4** and the bottom endpoint **25** of the via **21** may be shorter than a vertical distance between the substrate **4** and the bottom level **62***b* of the top FET **62**. The bottom level **62***b* of the top FET **62** may be a bottom level of the second source/drain region **52**. The via **21** may be laterally spaced apart from the stack of FETs **60**. Thus, the via **21** may be laterally spaced apart from the second source/drain region **52**.

[0093] The interconnect **31** may comprise a top part **30***p* of a trench **30** filled with electrically conductive material, the trench **30** extending from a bottom side **4***b* of the substrate **4** towards the first insulating layer **11**, wherein the top part **30***p* of the trench **30** comprises both the bottom side **51***b* of the first source/drain region **51** and the bottom endpoint **25** of the via **21**.

[0094] The interconnect **31** may extend, in the figure, laterally between the via **21** and the first source/drain region **51**. Thus, the via **21** may be electrically connected to the first source/drain region **51** by the interconnect **31**. In the figure, the interconnect **31** may extend laterally in the first direction, i.e. the X-direction. The interconnect **31** may extend vertically in the third direction, i.e. the Z-direction, such that the interconnect **31** connects at least to the bottom side **51***b* of the first source/drain region **51**. In the figure, the interconnect **31** connects additionally to part of a lateral side **51***a* of the first source/drain region **51**. In the figure, the interconnect **31** connects to the lower part of a lateral side **51***a* of the first source/drain region **51**.

[0095] The semiconductor device **1** may, as shown in FIG. **1**, further comprise a second insulating layer **12** below the electrically conductive material of the filled top part **30***p* of the trench **30**; and a backside power delivery network **13** below the second insulating layer, wherein the second insulating layer **12** may electrically insulate the backside power delivery network **13** from the electrically conductive material of the filled top part **30***p* of the trench **30**.

[0096] As seen in the figure, the second insulating layer **12** may fill the part of the trench **30** which is not filled with electrically conductive material.

[0097] It should be understood that the semiconductor device **1** may, alternatively, be implemented without a second insulating layer **12** and/or without a backside power delivery network **13**.

[0098] The semiconductor device **1** may, as shown in FIG. **1**, further comprise a MINT layer **70** comprising MINT lines **71** for electrically connecting the stack of transistors **60** to other electrical components, e.g. other stacks of transistors. The MINT layer **70** may be the first horizontal metal layer in the standard cell comprising the CFET. Accordingly, the MINT lines **71** may be metal lines. In FIG. **1**, the first source/drain region **51** is connected to a MINT line **71** through the via **21** and the interconnect **31**. Similarly in FIG. **1**, the second source/drain region **52** is connected to another MINT line **71** through another via **121** and another interconnect **131**.

[0099] In the following discussion, a method for forming a semiconductor device **1** is described in conjunction with FIGS. **2***a-d* to **5***a-b*. These figures show cross-sectional views of the semiconductor device **1**, during production, corresponding to the cross-sectional view of FIG. **1**.

[0100] FIGS. **2***a-d* to **5***a-b* all show formation of a semiconductor device **1** wherein the first source/drain region **51** and the second source/drain region **52** as well as connections to the second source/drain region **52**, by via **121** and interconnect **131**, are already in place. However, it should be understood that the method may comprise forming the features. In particular, the method may comprise: forming source/drain regions **51**, **52** of the bottom **61** and top **62** FET by epitaxial growth. The epitaxial growth may be performed before filling the hole **20** with electrically conductive material and filling the top part **30***p* of the trench **30** with electrically conductive material.

[0101] FIG. **2***a-d* illustrates the formation of a semiconductor device **1** wherein hole-etching and hole-filling are performed from the frontside and trench-etching and trench-filling are performed from the backside. The steps of FIG. **2***a-d* may be performed in the order indicated by the arrows between the figures.

[0102] FIG. **2***a* illustrates the semiconductor device **1** after forming the stack of FETs **60**, forming the first insulating layer **11**, and etching a hole **20** into the first insulating layer **11**. The hole **20** may be laterally spaced apart from the stack of FETs **60**, the hole **20** may extend between a top endpoint **26** and a bottom endpoint **25**, the top endpoint **26** may be arranged at a top side **11***t* of the first insulating layer **11**, the bottom endpoint **25** may be arranged at a level below a bottom level **62***b* of the top FET **62**. In the figure, the hole **20** is etched from the top side **11***t* of the first insulating layer **11**, i.e. frontside hole-etching. A characteristic of a hole **20** etched by frontside-etching may be that it tapers downwards, i.e. the width of the hole **20** at the top endpoint **26** may be larger than the width of the hole **20** at the bottom endpoint **25**.

[0103] FIG. **2***b* illustrates the semiconductor device **1** after filling the hole **20** with electrically conductive material such that the electrically conductive material of the filled hole **20** forms a via **21**.

[0104] FIG. **2***c* illustrates the semiconductor device **1** after a trench **30** being etched, from a bottom side **4***b* of the substrate **4** towards the first insulating layer **11**, such that a top part **30***p* of the trench **30** comprises both a bottom side **51***b* of the first source/drain region **51** and the bottom endpoint **25**. In the figure, the trench has been etched past the bottom side **51***b* of the first source/drain region **51** such that the top part **30***p* of the trench **30** comprises also part of a lateral side **51***a* of the first

source/drain region **51**, in addition to the bottom side **51***b* of the first source/drain region **51**.
[0105] In the figure, the trench **30** is etched from the bottom side **4***b* of the substrate **4** towards the first insulating layer **11**, i.e. backside trench-etching. A characteristic of a trench **30** etched by backside-etching may be that it tapers upwards, i.e. the width of the bottom of the trench **30** may be larger than the width of the top of the trench **30**.
[0106] FIG. **2***c* further illustrates that a MINT layer **70** may be formed on top of the first insulating layer **11** before etching the trench **30**. For example a MINT line **71** may be formed on top of the via **21**.
[0107] FIG. **2***d* illustrates the semiconductor device **1** after the top part **30***p* of the trench **30** being filled with electrically conductive material, such that the electrically conductive material of the filled top part **30***p* of the trench **30** electrically connects the bottom side **51***b* of the first source/drain region **51** to the bottom endpoint **25** of the via **21**, whereby the filled top part **30***p* of the trench **30** forms an interconnect **31**.
[0108] In the following description, backside hole-filling will be discussed in conjunction with FIGS. **3***a-d* and FIG. **4***a-d*.
[0109] FIG. **3***a-d* illustrate that a hole **20** etched from the frontside may be filled using backside hole-filling, e.g. by use of a sacrificial material **29**. The steps of FIG. **3***a-d* may be performed in the order indicated by the arrows between the figures.
[0110] FIG. **3***a* illustrates the semiconductor device **1** after forming the stack of FETs **60**, forming the first insulating layer **11**, etching a hole **20** into the first insulating layer **11**, and depositing a sacrificial material **29** in the hole **20**.
[0111] FIG. **3***a* further illustrate that a MINT layer **70** may be formed on top of the first insulating layer **11** before etching the trench **30**. For example a MINT line **71** may be formed on top of the sacrificial material **29**.
[0112] FIG. **3***b* illustrates the semiconductor device **1** after a trench **30** being etched, from the bottom side **4***b* of the substrate **4** towards the first insulating layer **11**, such that the top part **30***p* of the trench **30** comprises both the bottom side **51***b* of the first source/drain region **51** and the bottom endpoint **25**. The bottom endpoint **25** may herein be marked by the bottom end of the sacrificial material **29**.
[0113] FIG. **3***c* illustrates the semiconductor device **1** after removing the sacrificial material **29** in the hole **20** from the bottom side **4***b* of the substrate **4**. The sacrificial material **29** may be removed by etching, e.g. by selective etching.
[0114] FIG. **3***d* illustrates the semiconductor device **1** after filling the hole **20** with electrically conductive material and filling the top part **30***p* of the trench **30** with electrically conductive material. The hole **20** and the top part **30***p* of the trench **30** may be filled with electrically conductive material simultaneously, e.g. in one single deposition.
[0115] FIG. **4***a-d* illustrate that a hole **20** etched from the backside may be filled using backside hole-filling. The steps of FIG. **4***a-d* may be performed in the order indicated by the arrows between the figures.
[0116] FIG. **4***a* illustrates the semiconductor device **1** after forming the stack of FETs **60** and forming the first insulating layer **11**. At this point neither the hole **20** nor the trench **30** has been etched.
[0117] FIG. **4***a* further illustrates that a MINT layer **70** may be formed on top of the first insulating layer **11** before etching the hole **20** and before etching the trench **30**. For example a MINT line **71** may be arranged on top of the first insulating layer **11** such that the via **21**, when formed, connects to the MINT line **71**.
[0118] FIG. **4***b* illustrates the semiconductor device **1** after a trench **30** being etched, from the bottom side **4***b* of the substrate **4** towards the first insulating layer **11**, such that the top part **30***p* of the trench **30** comprises the bottom side **51***b* of the first source/drain region **51**.
[0119] FIG. **4***c* illustrates the semiconductor device **1** after etching the hole **20** from the bottom side

4*b* of the substrate **4**. The hole **20** may herein be etched from a top end of the trench to the top side **11***t* of the first insulating layer **11**, i.e. from the bottom endpoint **25** to the top endpoint **26**. Thus, the top part **30***p* of the trench **30** comprises the bottom endpoint **25**. A characteristic of a hole **20** etched by backside-etching may be that it tapers upwards, i.e. the width of the hole **20** at the top endpoint **26** may be smaller than the width of the hole **20** at the bottom endpoint **25**.
[0120] FIG. **4***c* illustrates the semiconductor device **1** after filling the hole **20** with electrically conductive material and filling the top part **30***p* of the trench **30** with electrically conductive material. The hole **20** and the top part **30***p* of the trench **30** may be filled with electrically conductive material simultaneously, e.g. in one single deposition.
[0121] FIG. **5***a-b* illustrate that parts of a hole **20** may be etched using an etchstop. The steps of FIG. **5***a-b* may be performed in the order indicated by the arrow between the figures.
[0122] In FIG. **5***a-b*, the first insulating layer **11** comprises a first sublayer **111**, a second sublayer **112** stacked on top of the first sublayer **111**, and a third sublayer **113** stacked on top of the second sublayer **112**, the second sublayer **112** being arranged between the bottom **61** and top **62** FETs, the second **112** and third **113** sublayers comprising different insulating materials. The first sublayer **111** may comprise SiO.sub.2. The second sublayer **112** may act as an etchstop layer. The second sublayer **112** may comprise Si.sub.3N.sub.4. The third sublayer **113** may comprise SiO.sub.2.
[0123] FIG. **5***a* illustrates the semiconductor device **1** after selectively etching a first part **27** of the hole **20** through the third sublayer **113** down to the second sublayer **112**. An example of a dry etch that selectively etches a third sublayer **113** of SiO.sub.2 but does not significantly etch a second sublayer **112** of Si.sub.3N.sub.4 is a fluorine based plasma etch. An example of a wet etch that selectively etches a third sublayer **113** of SiO.sub.2 but does not significantly etch a second sublayer **112** of Si.sub.3N.sub.4 is hydrofluoric acid.
[0124] FIG. **5***b* illustrates the semiconductor device **1** after forming a second part **28** of the hole **20** by expanding the first part **27** of the hole **20** through the second sublayer **112**. Thus, the second part **28** of the hole **20** may be formed by etching the second sublayer **112**. Either a selective or a non-selective etch may be used.
[0125] While some embodiments have been illustrated and described in detail in the appended drawings and the foregoing description, such illustration and description are to be considered illustrative and not restrictive. Other variations to the disclosed embodiments can be understood and effected in practicing the claims, from a study of the drawings, the disclosure, and the appended claims. The mere fact that certain measures or features are recited in mutually different dependent claims does not indicate that a combination of these measures or features cannot be used. Any reference signs in the claims should not be construed as limiting the scope.

Claims

1. A method for forming a semiconductor device, comprising: forming a stack of field effect transistors, FETs, on top of a substrate, the stack of FETs comprising a bottom FET and a top FET, the top FET being stacked on top of the bottom FET, the bottom FET comprising at least a first source/drain region; forming a first insulating layer, the first insulating layer laterally surrounding the stack of FETs; etching a hole into the first insulating layer, the hole being laterally spaced apart from the stack of FETs, the hole extending between a top endpoint and a bottom endpoint, the top endpoint being arranged at a top side of the first insulating layer, the bottom endpoint being arranged at a level below a bottom level of the top FET; filling the hole with electrically conductive material such that the electrically conductive material of the filled hole forms a via; etching, from a bottom side of the substrate towards the first insulating layer, a trench, such that a top part of the trench comprises both a bottom side of the first source/drain region and the bottom endpoint; and filling the top part of the trench with electrically conductive material, such that the electrically conductive material of the filled top part of the trench electrically connects the bottom side of the first source/drain region to the bottom endpoint of the via, whereby the filled top part of the trench forms an interconnect.

2. The method of claim 1, wherein the top part of the trench comprises also at least part of a lateral side of the first source/drain region, such that the electrically conductive material of the filled top part of the trench electrically connects also the lateral side of the first source/drain region to the via.

3. The method of claim 1, wherein a vertical distance between the bottom FET and the top FET is smaller than 50 nm.

4. The method of claim 1, the method further comprising: forming a second insulating layer below the electrically conductive material of the filled top part of the trench; and forming a backside power delivery network below the second insulating layer, wherein the second insulating layer electrically insulates the backside power delivery network from the electrically conductive material of the filled top part of the trench.

5. The method of claim 1, wherein the electrically conductive material filling the top part of the trench is a metal.

6. The method of claim 1, wherein forming the stack of FETs comprises: forming source/drain regions of the bottom FET and top FET by epitaxial growth; wherein filling the hole with electrically conductive material and filling the top part of the trench with electrically conductive material are performed after forming the source/drain regions of the bottom FET and top FET.

7. The method of claim 1, wherein etching the hole comprises etching at least part of the hole from a top side of the first insulating layer.

8. The method of claim 7, wherein the first insulating layer comprises a first sublayer, a second sublayer stacked on top of the first sublayer, and a third sublayer stacked on top of the second sublayer, the second sublayer being arranged between the bottom and top FETs, and the second and third sublayers comprise different insulating materials; wherein forming the hole comprises: selectively etching a first part of the hole through the third sublayer down to the second sublayer; and forming a second part of the hole by expanding the first part of the hole through the second sublayer.

9. The method of claim 7, the method further comprising: depositing a sacrificial material in the hole before etching the trench; and removing the sacrificial material in the hole after etching the trench, by etching, from the bottom side of the substrate, the sacrificial material; wherein filling the hole with electrically conductive material is performed by filling, from the bottom side of the substrate, the hole with electrically conductive material after removing the sacrificial material.

10. The method of claim 1, wherein etching the hole is performed after etching the trench and comprises: etching the hole from the bottom side of the substrate; wherein filling the hole with electrically conductive material is performed from the bottom side of the substrate.

11. A semiconductor device, comprising: a stack of field effect transistors, FETs, arranged on top of a substrate, the stack of FETs comprising a bottom FET and a top FET, the top FET being stacked on top of the bottom FET, the bottom FET comprising at least a first source/drain region; a first insulating layer, the first insulating layer laterally surrounding the stack of FETs; a via comprising a hole filled with electrically conductive material, the hole being laterally spaced apart from the stack of FETs, the hole extending between a top endpoint and a bottom endpoint, the top endpoint being arranged at a top side of the first insulating layer, the bottom endpoint being arranged at a level below a bottom level of the top FET; and an interconnect comprising a top part of a trench filled with electrically conductive material, the trench extending from a bottom side of the substrate towards the first insulating layer, wherein the top part of the trench comprises both a bottom side of the first source/drain region and the bottom endpoint of the via, the electrically conductive material of the filled top part of the trench electrically connecting the bottom side of the first source/drain region to the bottom endpoint of the via.

12. The semiconductor device of claim 11, wherein the top part of the trench comprises also at least part of a lateral side of the first source/drain region, such that the electrically conductive material of

the filled top part of the trench electrically connects also the lateral side of the first source/drain region to the via.
13. The semiconductor device of claim 11, wherein a vertical distance between the bottom FET and the top FET is smaller than 50 nm.
14. The semiconductor device of claim 11, further comprising: a second insulating layer below the electrically conductive material of the filled top part of the trench; and a backside power delivery network below the second insulating layer, wherein the second insulating layer electrically insulates the backside power delivery network from the electrically conductive material of the filled top part of the trench.
15. The semiconductor device of claim 11, wherein the electrically conductive material filling the top part of the trench is a metal.
16. The semiconductor device of claim 11, wherein the first source/drain region is formed by epitaxial growth.
17. The semiconductor device of claim 11, wherein the hole was etched at least in part from a top side of the first insulating layer.
18. The semiconductor device of claim 17, wherein the first insulating layer comprises a first sublayer, and a second sublayer stacked on top of the first sublayer.
19. The semiconductor device of claim 18, further comprising: a third sublayer stacked on top of the second sublayer, wherein the second sublayer is between the bottom and top FETs.
20. The semiconductor device of claim 19, wherein the second and third sublayers comprise different insulating materials.